Bear Spray Formulation Analysis: Reverse-Engineering Counter Assault

To qualify for an EPA B660 "Me-Too" registration pathway, the Paws Off bear spray formulation must be "substantially similar" to an existing registered product. Based on the provided Counter Assault EPA registration (EPA Reg. No. 55541-2) [1] and the EPA Market Characterization of the U.S. Defense Spray Industry [2], we can reverse-engineer the likely composition of the 98.0% "Other Ingredients" (inerts) in Counter Assault's formula.

Active Ingredient Baseline

The Counter Assault label explicitly states the active ingredient concentration:

- **Active Ingredient:** Capsaicin and related capsaicinoids (Derived from Oleoresin of Capsicum) — **2.0%** [1]
- **Other Ingredients: 98.0%** [1]

To match this, Paws Off must use exactly 2.0% capsaicinoids. The remaining 98.0% consists of three functional categories of inert ingredients: the propellant, the carrier/solvent, and the emulsifier/surfactant [2] [3].

Propellant Analysis (Estimated 60–80% of formula)

The propellant is the most critical inert ingredient, as it dictates the spray distance, volume, and flammability. Bear sprays require a "fog" delivery system that can project 30–40 feet [1] [2].

Historically, the defense spray industry relied on **HFC-134a** (1,1,1,2-Tetrafluoroethane) because it is non-flammable, has a high vapor pressure (666 kPa at 25°C), and a low boiling point (-26.1°C), allowing it to perform in freezing conditions [2]. However, the EPA report notes that in 2019, Counter Assault introduced a new bear deterrent that uses a propellant *other* than HFC-134a, and their product is now labeled as "Extremely Flammable" [1] [2].

Given the EPA SNAP (Significant New Alternatives Policy) approved list and the flammability warning, the most likely propellants used in the current Counter Assault formulation are:

1. **HFO-1234ze (1,3,3,3-Tetrafluoropropene):** This is the primary next-generation replacement for HFC-134a. While the EPA report notes it has a lower vapor pressure (reducing spray distance by ~35% in early tests) and formulation stability issues [2], recent patents (such as Honeywell's WO2018023077A1) have solved these issues by pairing HFO-1234ze with specific co-solvents like trans-HFO-1233zd [3].

2. **Hydrocarbon Blends (Propane/Isobutane/Butane):** Because the Counter Assault label now carries an "Extremely Flammable" warning [1], it is highly probable they transitioned to a hydrocarbon propellant blend (e.g., A-46 or A-70). Hydrocarbons provide excellent vapor pressure for the required 40-foot range and are significantly cheaper than HFCs or HFOs, though they introduce flammability risks.

Recommendation for Me-Too: If targeting the current Counter Assault registration, a **Propane/Isobutane blend** is the most likely propellant candidate to match their "Extremely Flammable" profile while achieving the 40-foot range.

Carrier/Solvent Analysis (Estimated 15–35% of formula)

Oleoresin Capsicum (OC) is a thick, hydrophobic oil that must be thinned into a liquid state to be aerosolized. The solvent must be EPA List 4B approved (Inert Ingredients of Minimal Concern) [4].

Common EPA-approved solvents for OC formulations include:

- **Isopropyl Alcohol (Isopropanol):** Highly effective at dissolving OC resin, evaporates quickly, and is EPA List 4B approved. It contributes to the flammability of the overall aerosol.
- **Propylene Glycol / Polyethylene Glycol:** Often used as a co-solvent to stabilize the mixture and prevent the nozzle from clogging. It is EPA List 4B approved (CAS 57-55-6) [4].
- **Water:** Some formulations use a water-based carrier to reduce costs and flammability, but this requires heavy emulsification. Given Counter Assault's flammability warning, a heavy water base is less likely than an alcohol/glycol base.

Recommendation for Me-Too: A primary solvent of **Isopropyl Alcohol** mixed with a small amount of **Propylene Glycol** is the industry standard for flammable OC aerosols.

Emulsifier/Surfactant Analysis (Estimated 1–5% of formula)

Because OC oil, the solvent, and the liquefied propellant must remain in a stable solution (or emulsion) inside the canister, a surfactant is required to prevent separation [3].

Common EPA-approved emulsifiers include:

- **Polysorbate 80 (Tween 80):** A standard, highly effective non-ionic surfactant used to emulsify oils into water/alcohol mixtures.
- **Sodium Xylene Sulfonate:** An ionic surfactant noted in recent defense spray patents for stabilizing OC with newer propellants [3].

Recommendation for Me-Too: Polysorbate 80 is the safest bet for a legacy formulation match.

Proposed "Me-Too" Formulation Candidate

To achieve a "substantially similar" formulation to Counter Assault (EPA Reg. No. 55541-2) and qualify for the B660 PRIA pathway, we propose the following baseline formulation for your regulatory consultant to verify against the EPA Confidential Statement of Formula (CSF) database:

Component	Function	Proposed Ingredient	Estimated % by Weight
Active Ingredient	Irritant	Oleoresin Capsicum (yielding 2.0% capsaicinoids)	~2.0%
Propellant	Aerosolization / Range	Propane / Isobutane Blend (A-70)	65.0% - 75.0%
Primary Solvent	OC Carrier	Isopropyl Alcohol	15.0% - 25.0%
Co-Solvent	Stabilizer / Anti-clog	Propylene Glycol	2.0% - 5.0%
Emulsifier	Mixture Stability	Polysorbate 80	1.0% - 3.0%

Note: This formulation will trigger the "Extremely Flammable" physical hazard warning, which perfectly aligns with the current Counter Assault label requirements [1].

Next Steps

1. **Consultant Verification:** Provide this proposed ingredient list to ACTA (or your chosen regulatory consultant). They can cross-reference these specific List 4B inerts against known CSFs for 55541-2 to confirm if this exact ratio qualifies as "substantially similar."
2. **Lab Prototyping:** Once the consultant confirms the target CSF, work with your aerosol filler to mix a prototype using these exact percentages to ensure it achieves the required 30–40 foot spray distance and 9–10 second empty time [1].

References

[1] U.S. Environmental Protection Agency. (2026). Pesticide Registration Improvement Act (PRIA) Labeling Amendment: Counter Assault Bear Deterrent (EPA Reg. No. 55541-2).

[2] ICF. (2021). Market Characterization of the U.S. Defense Spray Industry. Prepared for the U.S. Environmental Protection Agency.
[3] Honeywell International Inc. (2018). Propellant and Self-Protection Compositions. WO2018023077A1.
[4] U.S. Environmental Protection Agency. Inert Ingredients Ordered by CAS Number - List 4B.